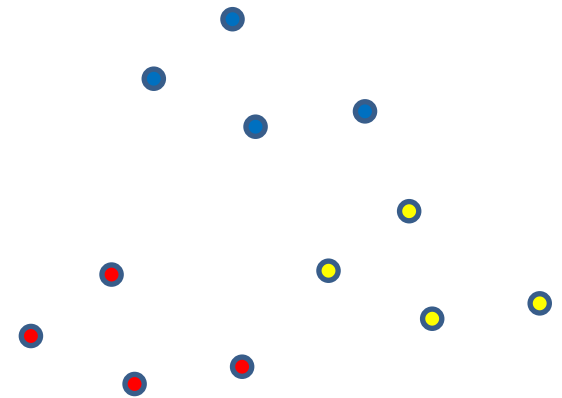


Regression

Lee-Ad Gottlieb

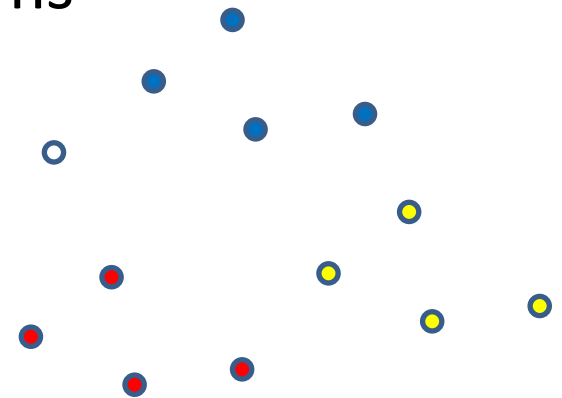
Multiclass

- So far, we've done **binary** classification
 - only two classes
 - Examples: $\{0,1\}$, $\{-1,1\}$, $\{-,+\}$, $\{\text{blue}, \text{red}\}$
- What about multiclass?
 - K classes
 - Ex: $\{1,2,3\}$, $\{\text{blue}, \text{red}, \text{yellow}\}$



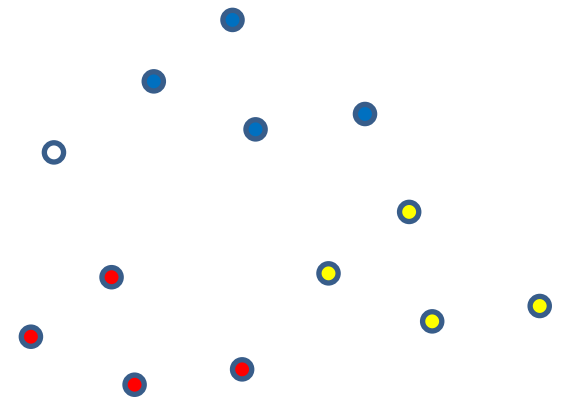
Multiclass

- What about multiclass?
 - K classes, {1,2,3}
- Reduce to binary **one-vs-one** problems
 - Is new point blue or yellow?
 - Is new point blue or red?
 - Is new point yellow or red?
- Winner: majority label
 - If there is one...
- This gives $O(k^2)$ subproblems (all pairs)



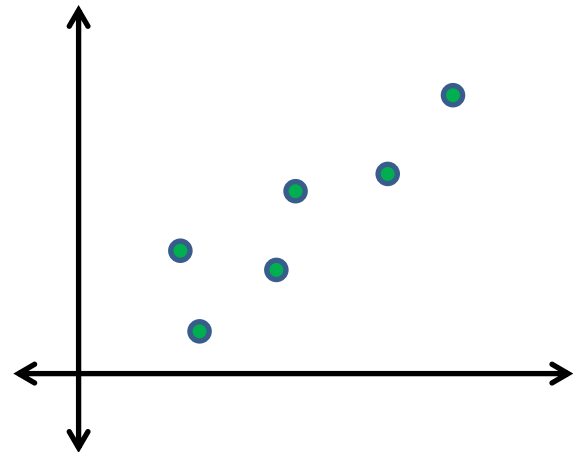
Multiclass

- What about multiclass?
 - K classes, {1,2,3}
- Reduce to binary **one-vs-all** problems
 - Is new point blue or not blue?
 - Is new point red or not red?
 - Is new point yellow or not yellow?
- Winner: whichever one wins...
 - If there is one...
- This gives only $O(k)$ subproblems
 - A different problem: **imbalanced data**
 - Possible solution: sample equal number from blue and not blue



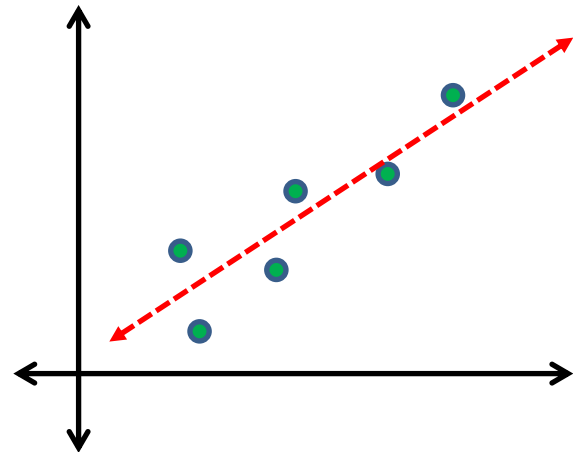
Regression

- Regression: Labels in real range
 - Example: $[0,1]$, $[-1,1]$, 0-120 kph, 0-200 kg.
 - More general: $\{0,M\}$ or $\{-M,M\}$ for some M
- Goal: Predict labels of unknown points



Regression

- Example:
 - In 1801, Giuseppe Piazzi observed the dwarf planet Ceres before its transit to the sun
 - Where will it appear after the transit?
- Gauss solved this using **linear regression**
- Legendre also used this method for astronomical prediction



Regression

- Problem: Predict labels of unknown points with continuous labels, for example $[0, M]$
- Notation:
 - Loss function: $L(h(x_i)) = |y_i - h(x_i)|^p$
 - $p \geq 1$.
 - Empirical risk: $\bar{r}(h) = \frac{1}{n} \sum_i L(h(x_i))$
 - True risk: $r(h) = E[L(h(x))]$ where $x \sim D$.
- Goal: Minimize true risk

Loss functions

- Suppose our data is just values

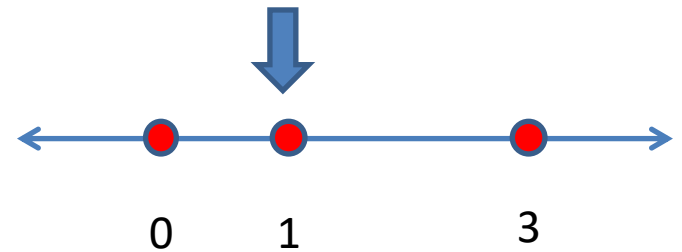
- Example: test scores

- Example: $Y: y_1=0, y_2=1, y_3=3$

- Goal: Minimize ℓ_1 loss

- $f = \operatorname{argmin}_z = \sum_i |z - y_i|$

- Solution: $z = \operatorname{median}(Y) = 1$

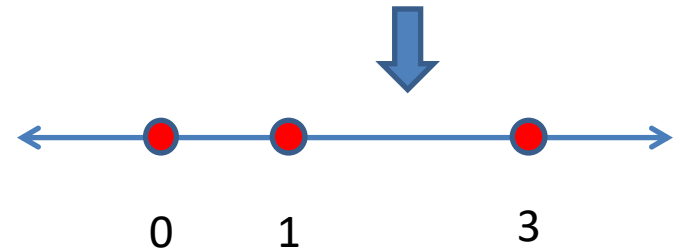


Loss functions

- Suppose our data is just values

- Example: test scores

- Example: $Y: y_1=0, y_2=1, y_3=3$



- Goal: Minimize ℓ_2 loss

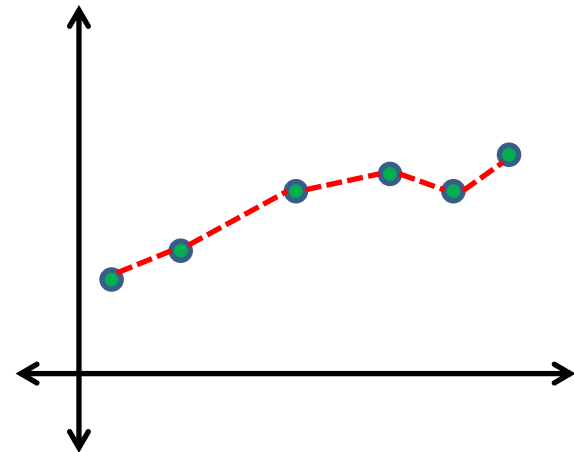
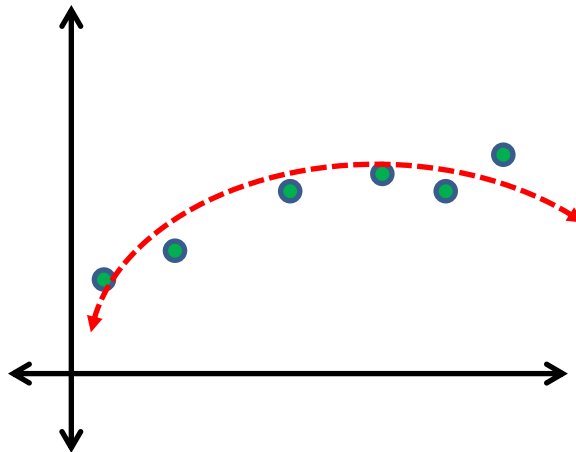
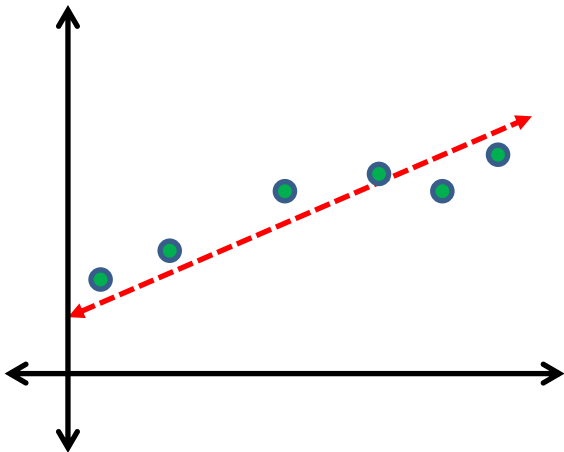
- $f = \operatorname{argmin}_z = \sum_i (z - y_i)^2$

- Solution: $z = \operatorname{mean}(Y) = 2$

- Proof: $f' = 2 \sum_i (z - y_i) = 0 \rightarrow z = \frac{1}{n} \sum_i y_i$

Regression

- What function should we try to fit our data to?
- Tradeoff:
 - Simpler function gives lower VC-dimension
 - Complex function can fit the data better



Linear regression

- Input
 - X: set of n points
 - Y: their labels
 - $H: R^d \rightarrow R$. Hypothesis space mapping points to labels. H is the infinite set of hyperplanes rooted at the origin, and h_w is one such hypothesis.
- Goal. Find hyperplane minimizing the **least-squares** loss

$$\operatorname{argmin}_{w \in R^d} \sum_{i=1}^n (w \cdot x_i - y_i)^2$$

Regression

- Generalization bounds for least squares
- Theorem: Let H be a finite hypothesis set, and let the range of labels be $[0, M]$. Let S be a sample of size n . Then for any $0 < \delta < 1$ with probability at least $1 - \delta$, the following holds:
- For all $h \in H$

$$r(h) \leq \bar{r}(h) + M \sqrt{\frac{\log|H| + \log(2/\delta)}{2n}}$$

Regression

- Generalization bounds for least squares
- Theorem: Let H have VC-dimension d , and let the range of labels be $[0, M]$. Let S be a sample of size n . Then for any $0 < \delta < 1$ with probability at least $1 - \delta$, the following holds:
- For all $h \in H$

$$r(h) \leq \bar{r}(h) + M \sqrt{\frac{O(d \log n + \log(1/\delta))}{n}}$$

Linear regression

- Restating the problem
 - Arrange the points in a $d \times n$ matrix X
 - Arrange the labels in a vector $Y \in R^n$
- The objective function is now
 - $f(w) = \|X^T w - Y\|^2$
 - $f'(w) = 2X(X^T w - Y) = 0$
 - $XX^T w = XY$
- Solution: $w = (XX^T)^{-1}XY$

Linear regression

- Solution: $w = (XX^T)^{-1}XY$
 - Question: What if XX^T is not invertible?
 - Use the **Pseudo-Inverse** of XX^T instead!

Linear regression

- Moore-Penrose pseudo-inverse of matrix A

- Denoted A^+
- Inverse properties: $AA^+A = A$ $A^+AA^+ = A^+$
- Hermitian properties: $(AA^+)^T = AA^+$ $(A^+A)^T = (A^+A)$

- Defined as:

$$A^+ = \lim_{\lambda \downarrow 0} (A^T A + \lambda I)^{-1} A^T = \lim_{\lambda \downarrow 0} A^T (A A^T + \lambda I)^{-1}$$

- Note:

- $A^T A + \lambda I$ and $A A^T + \lambda I$ are always invertible. ($A^T A$ and $A A^T$ positive semidefinite, adding λI makes them positive definite)
- The limit exists even if are not invertible.
- A^+ is not continuous in the entries of A: Consider $\epsilon \rightarrow \infty$

$$\begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix}^+ \quad \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\epsilon} \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^+ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

A

A^{-1}

A

A^+

Linear regression

- Upshot: Need to solve $XX^T w = XY$
 - If XX^T is invertible: $w = (XX^T)^{-1}XY$
 - Otherwise: $w = (XX^T)^+XY$
- Both $(XX^T)^{-1}$ and $(XX^T)^+$ can be computed in time $O(d^3)$
- Total runtime: $O(nd + d^3)$

Generalization bound

- With high probability, for all $h \in H$
 - $r(h) \leq \bar{r}(h) + O(\sqrt{d/n})$
 - (where $M=1$, and δ is fixed)
- What if the dimension is large?
 - Use large margin...
 - Like we did for classification

Generalization bound

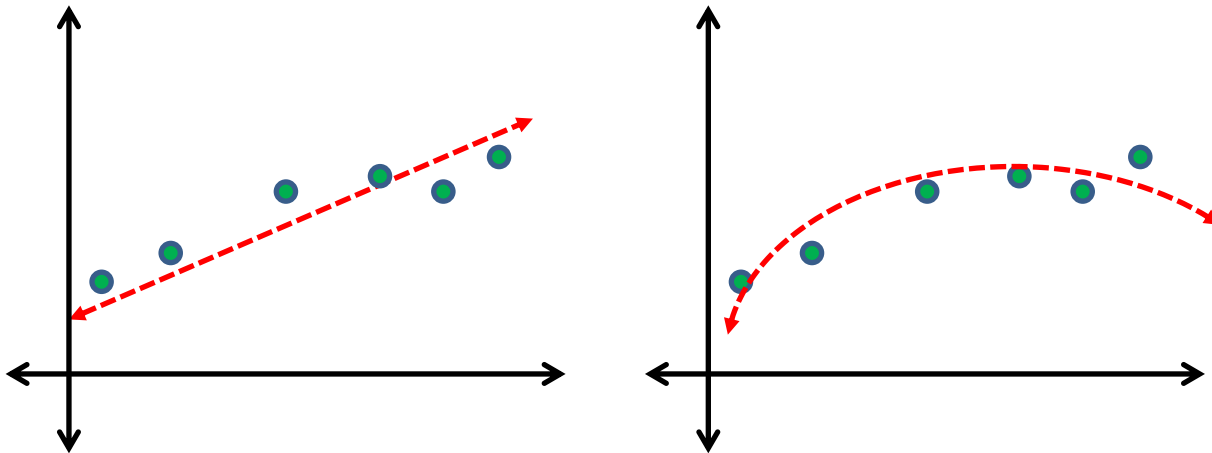
- Large margin bounds.
- If all three of the following hold:
 - Points: $\|x\| \leq R$ for all $x \in X$
 - Labels: $|y| \leq BR$ for all $y \in Y$
 - **Hypotheses**: $h_w \in H$ satisfy $\|w\| \leq B$
- Then with high probability for all $h_w \in H$
$$r(h) \leq \bar{r}(h) + O(B^2 R^2 / \sqrt{n})$$
- So the hypothesis needs to be **regularized**

Large-margin regression

- **Ridge regression**: regularized (large-margin) least squares
 - Analogous to large-margin SVM
 - $\text{Min}_{w \in R^d} [\lambda \|w\|^2 + \sum_{i=1}^n (w \cdot x_i - y_i)^2]$
- In matrix form:
 - $f(w) = \lambda \|w\|^2 + \|X^T w - Y\|^2$
 - $f'(w) = 2\lambda w + 2X(X^T w - Y) = 0$
 - $(XX^T + \lambda I)w = XY$
- Solution: $w = (XX^T + \lambda I)^{-1}XY$
 - (always invertible)

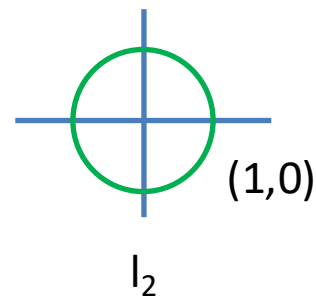
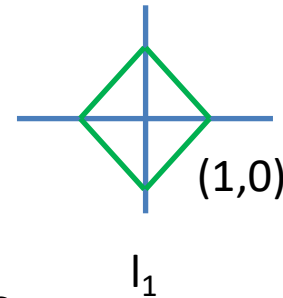
Non-linear function

- Ridge regression
 - How can we handle non-linear regressors?
 - Kernel function (Like in kernel-SVM)
 - $\text{Min}_{w \in R^d} [\lambda \|w\|^2 + \sum_{i=1}^n (w \cdot \phi(x_i) - y_i)^2]$



LASSO

- LASSO: ℓ_1 -regularized least squares:
 - $\min_{w \in \mathbb{R}^d} \lambda \|w\|_1 + \sum_i (w \cdot x_i - y_i)^2$
- Cons:
 - Requires solving quadratic program
 - Doesn't admit kernelizable solution
- Pros:
 - Encourages sparsity
 - Solutions often fall at extremal points
 - In ℓ_1 , these are sparse



More advanced models

- For time series:
 - ARMA – Autoregressive Moving Average
 - ARIMA – Autogressive Integrated Moving Average
- Some data is cyclic.
 - Example: Air travel, Corona patients

